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Practical LLM Applications in Pharmacoepidemiology and Pharmacovigilance

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Where this session fits

- **Earlier today:** how NLP and LLMs work, and the NLP lab
- **This session:**
    1. Common uses in pharmacovigilance
    2. Common uses in pharmacoepidemiology
    3. A framework (VALIDATE-AI) to read an LLM paper

    –> Is it potentially useful for your own research?

1. Common uses in pharmacovigilance

1 Case processing Hakim 2025

- Can a LLM translate case reports, with quality checks (e.g. inventing drug names)?
  - Common input data: Individual Case Safety Reports (ICSR) from spontaneous report systems, free notes in electronic health records

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3 Causality assessment Heckmann 2026 (also Abate 2025; Ha 2026)

- Can small local LLMs assign causality the way experts do?
- Causality Assessment Algorithm: The Naranjo and World Health Organization-Uppsala Monitoring Centre (WHO-UMC) algorithms

1. Lessons learnt: what would fix them?

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  - Larger, newer models handle clear-cut tasks better
  - Borderline “possible” cases stay hard even for the best models.

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- **Better system: quality checks built around the model**
  - Look up each drug name and adverse event term (e.g., Medical Dictionary for Regulatory Activities [MedDRA]), so invented or misspelled terms are caught
  - Input data quality matters: masked, incomplete reports will impact performance
  - ? For causality, use LLM to answer each algorithm question, and use code for scoring rules.

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- **Better workflow: how we test it and use it**
  - The reference standard should be fixed before anyone sees the model’s output.
  - Tasks work better when framed as a structured checklist than as an overall judgment.
  - ? A trained reviewer should check flagged cases and sign off every causality decision.

2. Common uses in pharmacoepidemiology

1 **Study design** Zhang 2026; Ompad 2026; Schneeweiss and Wang. Harmonize initiative 2026

- Can an LLM draft a study design and its code lists from a study title?
  - Common input data: study title or objectives; published protocols as the reference
  - Appraising studies for risk of bias (Okeke 2026);

2. Common uses in pharmacoepidemiology

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2. **Phenotyping** Owens 2026; Shao 2025; Gu 2025
   - Can an LLM measure exposures, outcomes and covariates?
     - Examples: social determinants of health, diabetes mellitus, medication discontinuation

3. **Predicting outcomes** Jiang 2023; Zhu 2026
   - Can a language model predict outcomes, such as readmission, from notes and codes?

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4. **Running whole studies with agents** Li 2026 (RWE-bench, preprint)
   - Can an AI agent re-run a published study from its protocol?

2. Lessons learnt: what would fix them?

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  - Larger models write good design text and read notes well
  - LLMs still leave codes out of code lists
  - May recall published studies they were trained on (data leakage) rather than design new ones

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- **Better system**
  - Build code lists from curated libraries and phenotype repositories
  - Map codes between coding systems automatically, so a valid code is not lost to a different convention (e.g., ICD-10-AM or ICD-10-CM)
  - Use the LLM where codes are not enough
  - **For AI agents**: log every step, and check automatically, e.g. no made-up patients, plausible cohort size, dates in the right order

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- **Better workflow**
  - Validate LLM-derived variables against chart review on our own data, as for any phenotype
  - Test on study protocol or publications written after the model's training data end
  - ? A person should review every code list and every cohort an agent builds

3. An LLM is another tool

- Read an LLM paper the way you read a validation study

3. A framework to read any LLM paper

Introduction
Aim

Methods
Data and model

Q1 Task
What exactly is the model asked to do?

Q2 Inputs
What does the model see? (e.g., type of data, pre-processing, language)

Q3 Model
Which LLM, which settings? (e.g., Exact version, size, prompt, temperature, runs)

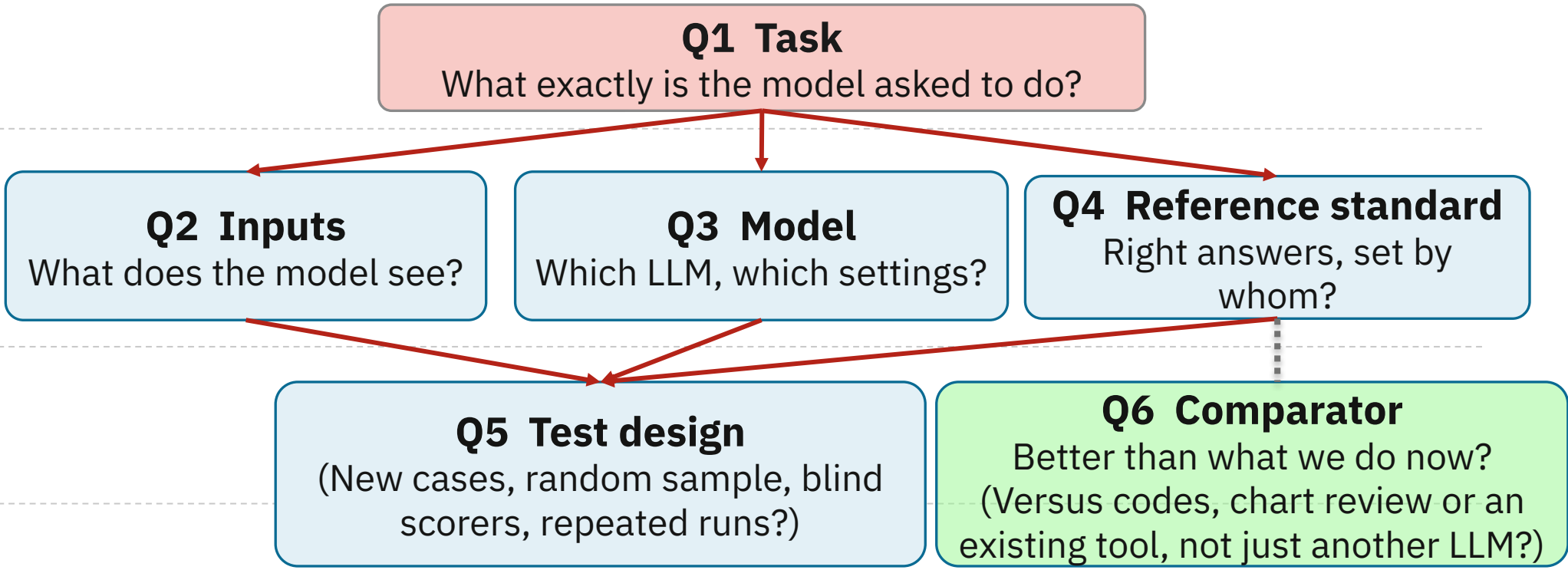
Q4 Reference standard
Right answers, set by whom? Before seeing the model's answers?)

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Evaluation



3. A framework to read any LLM paper

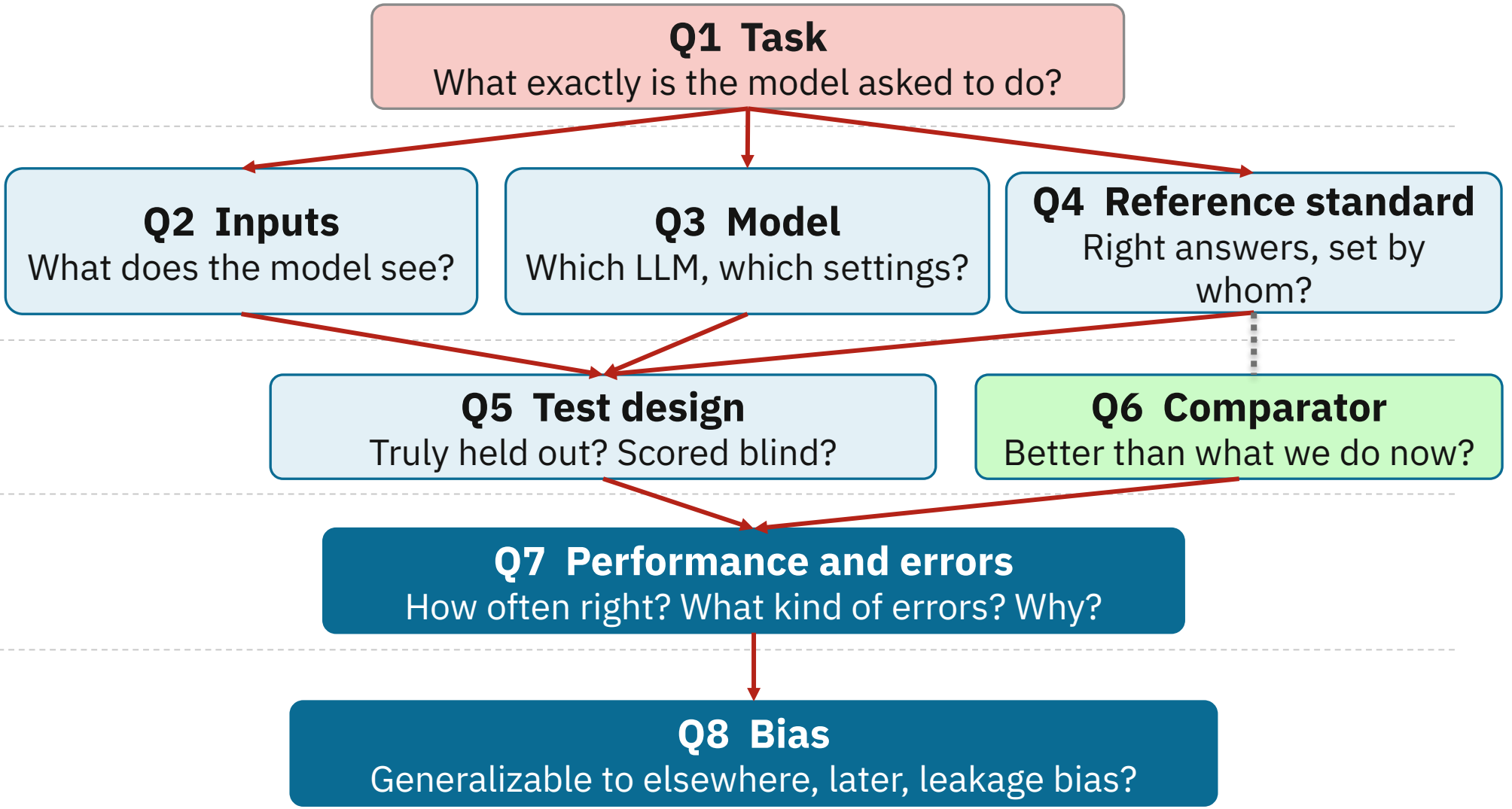
Introduction
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Results

Discussion
Limitations



Summary

- **Where LLMs help:** finding and organising evidence in notes and reports
- **Not ready on their own:** causality judgments and final code lists
- **What fixes most problems:** better systems and workflows around the model, more than a bigger model.
- **How to judge a paper:** ask the eight questions, starting with the reference standard and the test design.
- **Next, in Harvey's lab:** change the temperature and seed, and see whether the answers change.

Questions

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VALIDATE-AI: a framework for reading AI studies in epidemiology

A worked example

3. Deep dive - Zhang 2026: can an LLM draft a study design?

Pharmaceutical Research
https://doi.org/10.1007/s11095-026-04190-5

Check for updates

ORIGINAL RESEARCH ARTICLE

Employing General-Purpose and Biomedical Large Language Models with Advanced Prompt Engineering for Pharmacoepidemiologic Study Design

Xinyao Zhang[1] · Nicole Sonne Heckmann[1] · Manuela Del Castillo Suero[1] · Francesco Paolo Speca[1] · Maurizio Sessa[1]

3. Deep dive - Zhang 2026: can an LLM draft a study design?

Study overview:

1 Input

Only the title of a published study protocol

2 Models

- 4 LLMs
- 9 design questions,
- List the codes

3 Check

Compared with what the real protocol said (N=46 protocols)

4 Our question

Would we trust this draft for our own study?

https://github.com/madelsu/LLM-for-pharmacoepi-study-design

Question 1 of 8 · Task - What was the LLM asked to do?

- Ask LLMs to answer 9 design questions, in this order

1. Study design	2. Index date	3. Inclusion and exclusion criteria
4. Inclusion and exclusion assessment window	5. Exposure	6. Outcome
7. Follow-up	8. Covariates	9. Covariate assessment window

- Then, separately: list the codes for exposure and outcome (for example ICD, ATC, SNOMED CT)
  - The code-mapping step came after the 9 design questions
  - Its prompts refer to the answers for question 5 (exposure) and question 6 (outcome).

Question 2 of 8 · Inputs - What did the model see?

What they input into the model?

- Only the study title of each protocol
- No protocol document: no data source, population or analysis plan

Question 3 of 8 · Model - What was run?

What they did

- **Two large general models:** GPT-4o (version of 6 August 2024); DeepSeek-R1
- **Two small medical models:** Bio-Medical-Llama-3-8B (run in a compressed form, which can cost accuracy) and Qwen2-1.5B-medical

- **Two versions of prompts:**
    1. "least-to-most”: step by step, each answer passed on to the next question
    2. "active prompting”: one long, stand-alone prompt per question, with rules and a worked example (GPT-4o only)
    3. 42 model-and-prompt combinations were tested on 3 protocols, then tested 5 model-and-prompt combinations on all 46 protocols

Question 4 of 8 · Reference standard

What they did

- 46 published protocols, 2018 to 2024
  - DARWIN EU, EMA's data network (16); other EU studies in the HMA-EMA catalogue (15); FDA's Sentinel (15)
- Two pharmacoepidemiologists, wrote down each protocol's answers – the right answers
- **Reflection**
  - "Right" means "the same as the published protocol"
  - A different but sound choice could be scored as wrong

Question 5 of 8 · Test design - How were the answers checked?

What they did

- **Design answers:** GPT-4o scored each answer from 1 to 5 against the right answer
- **Spot check:** 2 authors checked a random 20% of GPT-4o's scores (347), blind to which model wrote each answer, and agreed with GPT-4o every time
- **Code lists:** compared code by code with the protocol's list; no LLM involved
- **Before testing:** the five model and prompt setups were picked from 42, using 3 protocols

Question 6 of 8 · Comparator - Better than what we do now?

What they did

- Models compared with each other
- Ways of asking (prompts) compared with each other, for GPT-4o
  - **Not compared:** a pharmacoepidemiologist drafting the same design, or existing code-list libraries

Reflection

- We learn which model drafts best, not whether drafting with an LLM saves time without losing quality

Question 7 of 8 · Performance - How well did it do?

What was scored	Result	Which setup
Design answers, scored 1 to 5	4 out of 5 on 8 of 9 questions (median)	GPT-4o, step by step; HMA-EMA catalogue protocols
Codes that matched the protocol	113 of 477 (24%)	Three setups combined
Of the codes the model listed, how many were right (precision)	52%	GPT-4o, step by step
Of the protocol's codes, how many the model found (recall)	28%	GPT-4o, step by step
Whole code list exactly right	**12.5%**	GPT-4o, step by step

Question 7 of 8 · Errors - Which errors happened?

What they found, in the 364 codes that did not match

- **Missing codes, 74%:** codes the protocol used that the model left out
- **Another coding system, 17%:** mostly real codes from the source terminology (SNOMED CT for conditions, RxNorm for drugs) instead of the OMOP concept IDs the protocol used.
- **Invented codes (wrong format or not in the code list):** only 1.4%
- ICD codes matched best

Reflection

- Missing codes do not look wrong; they quietly shrink the cohort
- Build code lists from curated sources. Use the LLM at most to suggest

Question 8 of 8 Bias - Will it work in another setting?

What the paper tested

- **Data sources:** EU and US protocols only (DARWIN EU, the HMA-EMA catalogue, Sentinel)
- **Input:** study titles only, could it be better?
- **Model version:** one dated version of GPT-4o; results may not hold for newer versions

Reflection

- Before relying on it, test it in our own setting: our data source, our coding system and our protocol style

Question 8 of 8 Bias - Did the model already know the answers?

What the paper shows

- **Public test material:** all 46 protocols are public -> may have been in the model's training data
  - **Timing:** GPT-4o's training data end in October 2023, and 29 of the 46 protocols are from 2023
  - **The authors:** memorisation “cannot be excluded”; they propose testing on protocols written after the cutoff

How it would change the results

- **Design answers:** a recalled answer matches the protocol, so relevance scores may be too high
- **Code lists:** recall would have helped here too, yet most lists were incomplete

Reflection

- **Ask of any LLM benchmark: could the test material have been in the model's training data?**

  -> Data leakage?